

Determinism soak + sustained loadtest — captured evidence (HEAD eb9e1c4)

Field	Value
Revision	1
Last modified	2026-06-10T16:45:00Z
Status	GREEN — §11.4.50 deterministic-consistency + sustained-load stability confirmed
HEAD	eb9e1c4 (on main)
Authority	Operator mandate 2026-06-10 (“most stable build by morning; zero risk, zero bluff”)

Deeper-than-single-run stability evidence for the overnight session: repetition (§11.4.50 — green must be deterministic, not a one-off) + a §11.4.118 rare-defect discovery probe + sustained-load confirmation.

File	What it proves
determinism_soak_x5.log	Full Go module <code>go test -race -count=5 ./... → SOAK_EXIT=0</code> ; every package ok on all 5 iterations (deterministic green).
deep_race_soak_x10.log	Concurrency-heavy pkgs <code>go test -race -count=10 ./internal/{api,store,deviceemu,fabric} → DEEPSOAK_EXIT=0</code> ; no rare race surfaced at 10× under the race detector.
dashboard_vitest_determinism_x3.txt	Dashboard Vitest run 3× consecutively → 58/58 every run (UI-tier determinism).
loadtest_soak.log	Black-box sustained load (ephemeral ota-server, 30s, c=64): 1,145,543 requests, 38,184 rps, p50 1.16ms / p90 3.77ms / p99 7.21ms, 0 non-2xx .

Honest notes (§11.4.6)

- The loadtest reports 61 “errors (no response)” out of 1,145,543 (0.005%). These are **client-side connection-level** events (no HTTP response received) under extreme local concurrency — non_2xx: 0 confirms the server returned a valid 2xx for **every** request it answered. Not a server defect; characteristic of local ephemeral-port / TCP-backlog pressure at ~38k rps. Recorded, not hidden.
- The ephemeral server was booted on :8080 (memory store) and torn down on exit (trap) — self-cleaning, no leak. No credential involved (healthz is unauthenticated).